COURSE NAME: PLANT PROTECTION

COURSE CODE: NR 430

- PATHOLOGY-
LECTURE 1:

1. INTRODUCTION

Plants cover the most area of the earth's living environment as trees, grasses, flowers, and so on. Plants play different important roles in the environment such as ecosystem balance and food supplement for animals and humans.

The study of plant pathogens belongs to the branch of biology known as **plant pathology**.

Diseases in plants are classified according to the two types of origins and these are **biotic and/or abiotic** origin. *Biotic (infectious)* diseases are developed owing to microbial infection, while abiotic (noninfectious) diseases are developed due to unfavorable environmental conditions, such as lack or excess of nutrients, moisture, light, and the presence of toxic chemicals in air or soil etc.

Plant pathology therefore focuses on the study of the organisms and the environmental factors that cause disease in plants; the mechanisms or methods by which these factors induce disease in plants and the methods of preventing or controlling disease aiming at reducing the damage it causes.

Essentially, plant pathology is correlated with other sciences such as entomology, bacteriology, mycology, virology, and weed science due to the visible effects of insects, bacteria, fungi, viruses/viroids, and weeds on plants, respectively

Terms and concepts in plant pathology:

- **Pathogen**: is defined as an organism causing disease to its host
- **Phytopathogens**: an organism which causes diseases in plants
- **Biotic diseases**: these are diseases caused by the pathogen (fungi, bacteria or viruses)
- **Abiotic diseases**: these are diseases which are caused by non-living agents like weather, soils, chemicals, or mechanical injuries. They are also referred to as non-infectious diseases.
- **parasitism:** an organism (parasite) lives on or in another organism (host) to obtain its required nutrition
- **Epiphytes**: any plant that grows upon another plant or object merely for physical support. Epiphytes have no attachment to the ground or other obvious nutrient source and are not parasitic on the supporting plants
- **Saprophytes:** An organism, especially a fungus or bacterium that lives on and gets its nourishment from dead organisms or decaying organic material. Saprophytes recycle organic material in the soil, breaking it down into in simpler compounds that can be taken up by other organisms.
- **pathogenicity**: this is the ability of an organism to cause a disease
- **Virulence:** is the disease producing power of an organism or ability of a pathogen or microbe to infect or damage a host. It can also be defined as the level of severity of the disease symptom.
- **Infectious diseases**: these are diseases that result from infection of a plant by a pathogen

- **Acquired resistance**: Plant resistance to disease activated after inoculation of the plant with certain microorganisms or treatment with certain chemical compounds.
- **Parasite**: An organism living upon or in another living organism (the host) and obtaining the food from the invading host.
- . **Biotroph**: A plant pathogenic fungus that requires living host cells i.e. an obligate parasite.
- **Hemibiotroph**: A plant pathogenic fungus that initially requires living host cells but after killing the host cell grows on the dead and dying cells.
- **Necrotroph**: A pathogenic fungus that kills the host and survives on the dying and dead cells.
- **Pathogenesis**: It is a process caused by an infectious agent (pathogen) when it comes in contact with a susceptible host.
- **Infection**: The initiation and establishment of a parasite within a host plant.
- . **Inoculum**: That portion of pathogen which is transferred to plant and cause disease.
- **Invasion**: The penetration and spread of a pathogen in the host.
- **Colonization**: The growth of a pathogen, particularly a fungus, in the host after infection is called colonization.
- **Inoculum potential**: The growth or threshold of fungus available for colonization at substratum (host).
- **Symptoms**: The external and internal reaction or alterations of a plant as a result of disease.
- **Incubation period**: The period of time between penetration of a pathogen to the host and the first appearance of symptoms on the plant.
- **Disease cycle**: The chain of events involved in disease development.
- **Disease syndrome**: The set of varying symptoms characterizing a disease are collectively called a syndrome.
- **Alternate host**: Plants not related to the main host of parasitic fungus, where it produces its different stages to complete one cycle (heteroecious).
- **Collateral host**: The wild host of same families of a pathogen is called as collateral host.
- **Predisposition**: The effect of one or more environmental factors which makes a plant vulnerable to attack by a pathogen.
- **Physiologic race**: One or a group of microorganisms similar in morphology but dissimilar in certain cultural, physiological or pathological characters.
- **Symbiosis**: A mutually beneficial association of two or more different kinds of organisms.
- **Mutualism**: Symbiosis of two organisms that are mutually helpful or that mutually support one another.
- **Antagonism**: The counteraction between organisms or groups of organisms.

- **Mutation**: An abrupt appearance of a new characteristic in an individual as a result of an accidental change in genes present in chromosomes.
- **Disease**: Any deviation in the general health, or physiology or function of plant or plant parts, is recognized as a disease.
- **Crop Damage**: It is defined as any reduction in the quality or quantity of yield or loss of revenue resulting from crop injury.
- **Deficiency:** Abnormality or disease caused by the lack or abnormal level of availability of one or more essential nutrient elements.

2. THE CONCEPT OF DISEASE IN PLANTS

Unlike human beings and animals, it is not known whether plants also feel pain or discomfort and plants do not speak or otherwise communicate with us, it is difficult to pinpoint exactly when a plant is diseased especially if there are no visible signs.

A plant is considered to be healthy or normal, if it's able to carry out its physiological functions to the best of its genetic potential.

i. The meristematic (cambium) cells of a healthy plant should **divide and differentiate**,
and different types of specialized cells (like the phloem)must be able to **absorb water and nutrients from the soil**; translocate these to all plant parts;

ii. a plant must be able to carry out **photosynthesis**, translocate, metabolize, or store the photosynthetic products;

iii. **and must be able to produce seed** or other reproductive organs for survival and multiplication.

Therefore, if the plant is unable to carry out most of these activities due to the presence of the pathogen or unfavorable environmental condition, then it is said to be **diseased**.

Plant disease, then, can be defined as the series of invisible and visible responses of plant cells and tissues to a pathogenic organism or environmental factor that result in adverse changes in the function, or integrity of the plant and may lead to partial impairment or death of plant parts or of the entire plant. The kinds of cells and tissues that become affected determine the type of physiological function that will be disrupted first. For example, infection of roots may cause roots to rot and make them unable to absorb water and nutrients from the soil.

The first step of plant disease treatment is observation of definite and clear symptoms on the plants. These symptoms give an initial indication for the type and cause of plant disease, which may end with the death

The infected part of the plant gives an indication of the type of plant disease, such as infected root which is usually correlated with root-rot disease.

Plant diseases can be classified and identified according to **several parameters** and some of them are:

i. disease symptoms,
ii. infected organ,
iii. infected plant type,
iv. and the type of phytopathogen.

Identifying the phytopathogen is considered to be more useful criterion used for plant disease classification, because it easily determines the disease cause, potential disease complications, and offer possible control methods.

Pathogenic microorganisms cause diseases in plants by;

i. Disturbing the metabolism of plant cells through enzymes, toxins, growth regulators, and other substances they secrete and by absorbing foodstuffs from the host cells for their own use.

ii. Some pathogens may also cause disease by growing and multiplying in the xylem or phloem vessels of plants, thereby blocking the upward transportation of water or the downward movement of sugars, respectively, through these tissues

Thousands of plant diseases exists with some affecting a lot of plant species while just a few are specific. It is important to note that a single plant species can also be affected by a hundred or more diseases.

LECTURE 2.

1. TYPES OF PLANT DISEASES AND BASIC PROCEDURES IN THE DIAGNOSIS OF PLANT DISEASES

The plant disease diagnosis depends on the exact determination of the disease cause. Generally, there are two plant disease causes: the **pathogens** and/or **environmental factors**. The former leads to **infectious** diseases, while the latter leads to **non-infectious** diseases.

The most common method of diagnosing a plant disease is by grouping them according to the symptoms they cause

i. do they cause root rots, wilts, leaf spots, blights, rusts, smuts,
ii. the plant organ they affect (root diseases, stem diseases, foliage diseases),
iii. orto the types of plants affected (field crop diseases, vegetable diseases, tree diseases, turf diseases etc.

1.1. Infectious diseases

There are wide range of phytopathogens which cause infectious plant diseases such as fungi, bacteria, viruses, viroids, and protozoa. The infectious disease means the ability of phytopathogen to transfer diseases from the infected plant to another healthy one and causes the same disease and the same symptoms. Most phytopathogens can inhabit the internal environment of plants; however, some others can live on the plant surface such as some fungi, bacteria, and parasitic higher plants

Infectious or biotic plant diseases

- Diseases caused by fungi
- Diseases caused by bacteria
- Diseases caused by parasitic higher plants and green algae
- Diseases caused by viruses and viroids
- Diseases caused by nematodes

Noninfectious, or abiotic plant diseases

- Diseases caused by too low or too high temperatures
- Diseases caused by lack or excess of soilmoisture
- Diseases caused by lack or excess of light
- Diseases caused by lack of oxygen
- Diseases caused by air pollution
- Diseases caused by nutrient deficiencies
- Diseases caused by mineral toxicities
- Diseases caused by soil acidity or alkalinity (soil pH)

1.1.1. Diseases caused by parasitic higher plants

Some plant diseases are developed due to growing of certain plants attached on or in other plants, where they take all required nutrients without benefit sharing; these plants are called parasitic higher plants. This abnormal relationship leads to weakness of healthy or host plant. The parasitic higher plants are usually found attached with the surface of the host plant, such as dodder, witchweed, and broomrape.

1.1.2. Diseases caused by nematodes

The nematodes are one of most common phytopathogens which have definite symptoms. These symptoms only appear in the infected site. The nematode infections in or on plants are widely distributed especially in proper environments such as moderate temperature and high humidity.

2.1.3. Diseases caused by fungi

Interestingly, there are two main types of fungi appearing on plants: pathogenic and saprophytic. The pathogenic fungi live in or on plant tissues and cause serious complications for the vital physiological functions of plants, while saprophytic ones live in or on dead tissues. Accordingly, the diagnosis of plant disease must be exactly carried out. The exact diagnosis and determination of fungi take place by microscopical examination to identify the mycelial morphological characteristics, whatever fruiting structures and spores. After complete identification for the fungus and the symptoms of plant disease, the latter should be compared with that reported in the reference. This study will exactly determine whether the fungus is a pathogen or a saprophyte. Although microscopical examination is an essential and effective method for fungal identification, it only sometimes cannot lead to exact identification due to the absence of fungal fruiting structures and spores on infected plant tissue.

Therefore, an alternative method must be used, such as using selective media for isolation, identification, or promotion of sporulation. On the other hand, some fungi need to be incubated under certain temperature, aeration, or light conditions to produce spores

2.1.4. Diseases caused by bacteria and mollicutes

The appearance of bacterial growth in or on plant tissues means that bacterial plant disease may be present, because saprophytes may be present. Therefore, accurate bacterial identification must be carried out by using microscopical examination and physiological parameter determination.

The selective media are essentially used in the bacterial identification to determine the bacterial genus and species in some cases. Moreover, the confirmatory test of bacterial pathogenicity may be carried out by inoculation of single pure bacterial colony in the healthy plant, reproducing the same symptoms that appeared on the infected one. Moreover, immunodiagnostic techniques or serodiagnostic assays can be used, such as agglutination and precipitation, fluorescent antibody staining, and enzyme-linked immunosorbent assay (ELISA).

There are several advantages for these techniques such as quite sensitivity, fairly specificity, rapid, and easy to perform.

The molecular biological techniques are also widely used.

Mollicutes ;

There are uncommon microorganisms called **mollicutes and** similar to bacteria. These microorganisms are very small where they must be examined by an electron microscope. Mollicutes have polymorphism(2 or more variants of a particular DNA sequence) and lack cell wall–like mycoplasma. These microorganisms habit the young phloem cells as a convenient host, and cause severe plant diseases such as plant stunting, yellowing or reddening of leaves, proliferation of shoots and roots, production of abnormal flowers, and eventual decline and death of the plant. Mollicutes cannot be cultured on nutrient media except for the genus *Spiroplasma*. Mollicutes can be diagnosed by several parameters, such as symptoms determination, grafting, transformation, microscopical examination, susceptibility to tetracyclines, and so on.

2.1.5. Diseases caused by viruses and viroids

There are distinctive types of plant diseases caused by viruses/viroids. These diseases have definite and clear symptoms, which easily support disease diagnosis and are considered main advantage. Apart from this advantage, some recent techniques are widely used for disease diagnosis and virus identification, such as virus transmission tests to specific host plants by sap inoculation, grafting, certain insect, nematode, fungus, and mite vectors.

Moreover, serodiagnostic tests are used for this purpose such as enzyme-linked immunosorbent assays (ELISA), gel diffusion tests, micro-precipitin tests, and fluorescent antibody staining. The

electron microscopy techniques as negative staining of virus particles in leaf dip or purified preparations are also used, as well as immune-specific electron microscopy.

On the other hand, there are more accurate techniques used for disease diagnosis and virus/viroid identification, such as electrophoretic tests and hybridization of commercially available radioactive DNA complementary to a certain virus DNA or RNA, or viroid RNA, with the DNA or RNA present in plant sap and attached to a membrane filter (immunoblot).

2.1.6. Diseases caused by more than one pathogen

Sometimes, some plants are exposed to co-infection by two or more pathogens, which lead to the same or different disease symptoms. Therefore, the differentiation and identification of these pathogens are very essential to exactly determine the disease cause. The differentiation and subsequently identification are carried out by all techniques that are mentioned above.

2. PARASITISM AND PATHOGENICITY

The term parasitism called on the state in which an organism (parasite) lives on or in another one (host) to obtain its required nutrition. Usually, parasitism is correlated with **pathogenicity**, which means the ability of an organism to cause a disease.

Parasitism in some cases leads to a benefit relationship called **symbiosis**, in which both plant and organism alternate the benefits, such as bacterial nodules in the roots of legume plants and the mycorrhizal infection of feeder roots of most flowering plants. In the case of parasitism-pathogenicity relationship, the plant is diseased with the appearance of different symptoms such as increased respiration, disintegration or collapse of cells, wilting, abscission, abnormal cell division and enlargement, and degeneration of specific components such as chlorophyll.

The most common plant pathogens are fungi, bacteria, mollicutes, parasitic higher plants, parasitic green algae, nematodes, protozoa, viruses, and viroids. These parasites cause serious plant diseases, because they have the ability to penetrate the plant tissues to feed and proliferate in it, and withstand the conditions in which the host lives. These pathogens are also called **obligate parasites** because they can only live in their living hosts.

On the other hand, there are certain pathogens such as most fungi and bacteria can live on either living or dead hosts and on various nutrient media, so they are called **nonobligatory parasites**. Some nonobligatory parasites can grow saprophytically on dead organic matter, and therefore called semi-biotrophs/**facultative saprophytes**. There is a type of a life called facultative parasitism, in which an organism grows saprophytically (necrotrophs); however, under certain conditions, they attack living plants and cause a disease; these parasites are called **facultative parasites**. The type or degree of parasitism does not affect the disease severity. For instance, many diseases caused by weakly parasitic pathogens are much more damaging to a plant than others caused even by obligate parasites.

3. LOSSES CAUSED BY PLANT DISEASES

i. Plant Diseases Reduce the Quantity and Quality of Plant Produce

The type and amounts of losses caused by plant diseases vary with the plant, the pathogen, the locality, the environment, the control measures practiced, and combinations of these factors. The quantity of loss may range from slight to 100%. Plants or plant products may be reduced in quantity whilst in the field , or during storage.

ii. Plant Diseases May Limit the Kinds of Plants and Industries in an Area

Plant diseases may limit the kinds of plants that can grow in a large geographic area due to the effects of the disease dominant in that particular location.

iii. Plant Diseases May Make Plants Poisonous to Humans and Animals

Some diseases are very bad and make plant products unfit for human or animal consumption by contaminating them with poisonous fruiting structures.

iv. Plant Diseases May Cause Financial Losses

Financial losses from plant diseases can arise in many ways as farmers spend a lot in estabslishing farms. Therefore, plant diseases can cause losses and a farmer can lose the entire field in certain instances.

v. Economic loses

A farmer can experience serious loses as a result of reduce quality and quantity of the produce.

LECTURE 3.

1. PLANT DISEASE EPIDEMIOLOGY

Epiphytology or Epidemiology of plant diseases is essentially a study of the rate of multiplication of a pathogen and spread of the disease caused by it in a plant population. Epidemiology deals with outbreaks and spread of diseases in a population.

2. **Importance of epidemiology**:
Knowledge of epidemiology is useful in forecasting of a disease and also for the management of a disease
Terms compound interest and simple interest diseases were given by **Vanderplank** (1963) in his book "Plant Disease Epidemics and control"

S.No.	Compound interest disease/ polycyclic	Simple interest disease/Monocyclic
1	Rate of increase of disease is mathematically analogous to compound interest in money (Interest is added periodically to the capital; interest gets interest)	Rate of increase of disease is mathematically analogous to simple interest in money (Interest is added only at the end; interest does not get interest)
2	Pathogen produces spores at rapid rate	Pathogen produces spores at very slow rate
3	Propagules disseminate by air	Propagules disseminate by soil or seed
4	Incubation period and sporulation period is short	Incubation period and sporulation period is long
5	There are several generations of the pathogen in the life of a crop	There is only one generation of thepathogen in the life of a crop
6	Ex: Rusts of cereals	Ex: Smuts of wheat, barley & sorghum

The plant disease means the occurrence of physiological disorder(s) due to biotic agents such as microbial infection and/or abiotic agents such as extreme environmental factors.

In order for the plant disease to occur, an interaction must happen between two components: the plant and disease cause, which leads to physiological disorders. The disease cause is either biotic agent or abiotic agent as mentioned above. Interestingly, the biotic agents lead to infectious diseases, which develop under suitable environmental conditions. Therefore, the infectious diseases (occurred by pathogens) are not developed under extreme environmental conditions. This means it was impossible to get infectious and noninfectious plant diseases at the same time. The abiotic agents (environmental factors) play an important and vital role in the disease development and severity or disease resistance.

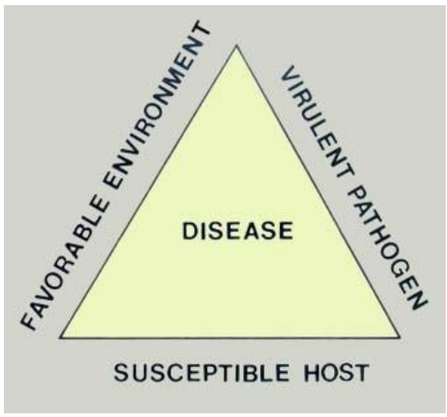
This matter depends mainly on **different factors**: the plant family, the plant age, plant genetic type, pathogen virulence race, pathogen inoculum size, and pathogen dormant state. Therefore, we can imagine the plant disease as a triangle, which is called "**disease triangle.**" The three sides of this triangle are the **plant**, **microorganisms**, and **the environmental factors**. The length of each side is proportional to the sum of the characteristics of the other two sides. For example, if the plant is resistant, the host side and the amount of disease would be small or zero, whereas if the plant is susceptible, the host side would be long and the potential amount of disease could be great .

2.1. Disease triangle

A plant becomes diseased in most cases when it is attacked by a pathogen or when it is affected by an abiotic agent. Therefore, in the first case, for a plant disease to occur, at least two components (plant and pathogen) must come in contact and must interact.

Additionally, a third component, namely a set of environmental conditions within a favorable range, must also occur for disease to develop.

The interactions of the three components of disease have often been visualized as a triangle and generally referred to as the "**disease triangle**." Each side of the triangle represents one of the three components.



I. Pathogen factors

- Virulence
- Inoculum type
- Dispersal potential
- Host range
- Alternation between hosts

II. Host plant factors

- Host resistance
- Age of the host plant
- Seasonal development
- Host uniformity

III. Environmental factors

- **Temperature:** Each pathogen has an optimum temperature for growth. In addition, different growth stages of fungi, such as the production of spores (reproductive units), their germination, and the growth of the mycelium (the filamentous main fungus body), may have slightly different optimum temperatures.

- **Light**
  Plants which receive adequate light will have less pathogen invesion as the conditions around them will be dry and physiological processes in the plants will be at their highest e.g photosynthesis.

- **Soil type**
  Certain pathogens are favoured by loam soils and others by clay soils. Usually, light soils that are low in organic matter are more preferred by pathogens.

- **Soil pH.**

A measure of acidity or alkalinity, remarkebly influences a few diseases. Most pathogensthrive well in acidic conditions, hence need for soil neutralization through the additional of lime. Some pathogens may not survive high soil p H levels.

- **Moisture**
  High or low soil moisture may be a limiting factor in the development of certain root rot diseases. High soil-moisture levels favour development of destructive water mold fungi,. Excessive watering of palnts is a common problem. Overwatering, by decreasing oxygen and raising carbon dioxide levels in the soil, makes roots more susceptible to root-rotting organisms.
- **Humidity**
  High humidity favours development of the great majority of leaf and fruit diseases caused by fungi, water molds, and bacteria. Moisture is generally needed for spore germination, the multiplication and penetration of bacteria, and the initiation of infection.

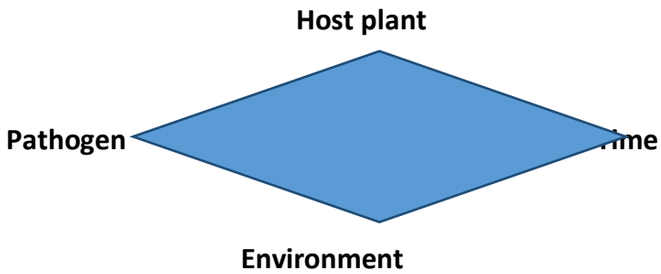
- **Air pollutant**

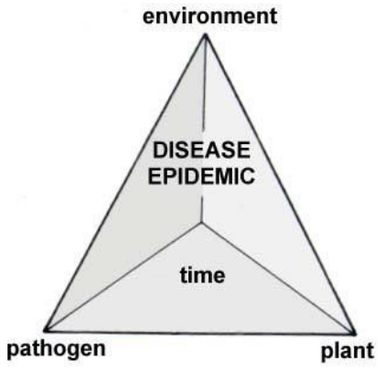
Pollution also influences disease development either by aiding pathogen development or in reducing their spread.

IV. Time

The **fourth factor** has been observed to have an influence in disease development is **'time'**.

Disease Pyramid: The disease triangle can be expanded to include two more components, time and humans. The amount of each of the three components of disease and their interaction in the development of the disease are affected by fourth component, time. Thus addition of time component to the disease triangle results into a **tetrahedron** or **disease pyramid**. The effect of time on disease development becomes apparent when we consider the importance of time of year, the duration and frequency of favourable temperature and rain, the time of appearance of the vector, the duration of the cycle of a particular disease. If the four components of disease pyramid could be quantified, its volume would be proportional to the amount of disease on a plant or in plant population. Humans affect disease development in various ways. They affect the type of plants grownin an area, their level of resistance, time of planting, density of planting, etc.





1.2 PATHOGENESIS

Pathogenesis is the stage of disease in which the pathogen is in intimate association with living host tissue. **Three** fairly distinct stages are involved:

Inoculation: transfer of the pathogen to the infection court, or area in which invasion of the plant occurs (the infection court may be the unbroken plant surface, a variety of wounds, or natural openings—e.g., stomata [microscopic pores in leaf surfaces], hydathodes [stomata-like openings that secrete water], or lenticels [small openings in tree bark].

Incubation: the period of time between the arrival of the pathogen in the infection court and the appearance of symptoms

Infection: the appearance of disease symptoms accompanied by the establishment and spread of the pathogen.

ROLE OF GROWTH REGULATORS IN PLANT PATHOGENESIS

Growth regulators

Growth regulators are of two types

i. Growth promoting substances and
ii. Growth inhibiting substances

Auxins, gibberellins and cytokinins are growth promoting substances, whereas, dormin, ethylene and abscissic acid are growth inhibiting substances.

The imbalance in growth promoting and growth inhibiting substances causes **hypertrophy** (excessive increase in cell size) and **atrophy** (decrease in cell size). Symptoms may appear as tumors, galls, knots, witches broom, stunting, excessive root branching, defoliation and suppression of bud growth.

1. Growth promoting substances:

a) Auxins: *Indole-3-acetic acid (IAA)* is the naturally occurring auxin. It is continuously produced in young meristematic tissue and moves rapidly to older tissues. If auxin concentration is more, its concentration is reduced by the enzyme, **IAA oxidase**.

***Functions*:** IAA regulates cell elongation and differentiation, also affects permeability of the membrane, increases respiration, and promotes synthesis of mRNA.

How disease is induced?

Increased IAA results in **hypertrophy** and decreased IAA results in **atrophy**. Increased IAA may be due to inhibition of IAA oxidase.

Ex: *Ralstonia solanacearum* (*Pseudomonas solanacearum*), the causal agent of wilt of Solanaceous plants, induces a 100 fold increase in IAA level in diseased plants. Increased plasticity of cell walls as a result of high IAA levels renders the pectin, cellulose and protein components of the cell wall more accessible to pathogen degradation. Increase in IAA levels may also inhibit lignifications of tissues.

Increased IAA levels have been reported in plants infected with the following pathogens *Phytophthora infestans* (Late blight of potato), *Ustilago maydis* (Maize smut), *Plasmodiophora brassicae* (Club root of crucifers), *Sclerospora graminicola* (Downy mildew of sorghum), *Agrobacterium tumefaciens* (Crown gall of apple),and *Meloidogyne* (Root knot nematode).

b) Gibberellins: First isolated from ***Gibberella fujikuroi*** (Conidial stage: *Fusarium moniliforme*), the causal agent of bakanae or foolish seedling disease of rice. Infected seedlings show abnormal elongation due to excessive elongation of internodes. Bestknown gibberellin is **Gibberellic acid**.

Functions: Cell elongation, stem and root elongation, promote flowering and growth of fruits. It also induces IAA synthesis. IAA and GA act synergistically. Ex: *Sclerospora sacchari*, the causal agent of downy mildew of sugarcane induces GA production.

c) Cytokinins: Kinetin was the first compound isolated from herring sperm DNA and does not occur naturally in plants. Cytokinins, such as **zeatin** and **isopentenyl adenosine** (IPA) have been isolated from plants

Functions: Cytokinins are necessary for cell growth and differentiation. It inhibits breakdown of proteins and aminoacids and thereby inhibit senescence and they have the capacity to direct the flow of aminoacids and other nutrients towards high cytokinin concentration. Cytokinin activity increases in club root, in crown galls and in rust infected bean leaves. Ex: Green islands are formed around infection in bean (*Phaseolus vulgaris*) leaves infected by *Uromyces phaseoli*.

2. Growth inhibiting substances

a) Ethylene (CH_2=CH_2): Ethylene exerts a variety of effects on plants, viz., chlorosis, leaf abscission, epinasty, stimulation of adventitious roots, fruit ripening and increased permeability of cell membranes.

Ex: Ethylene is involved in premature ripening of fingers in banana infected by *Pseudomonas solanacearum,* the causal agent of moko disease of banana. Ethylene was also detected in leaf epinasty symptom of the vascular wilt syndrome. Ex: *Fusarium oxysporum f.sp. lycopersici* (Wilt in tomato).

b) Abscissic acid: It exerts dormancy in seeds, closure of stomata, inhibition of seed germination and growth and stimulated germination of fungal spores. It is one of the factors involved in stunting of plants.

c) Dormin / Abscissin II: Dormin induces dormancy by converting developing leaf primordia of a bud into bud scales. It acts as an antagonist of gibberellins and masks the effect of IAA. However, the exact role of dormin is not known.

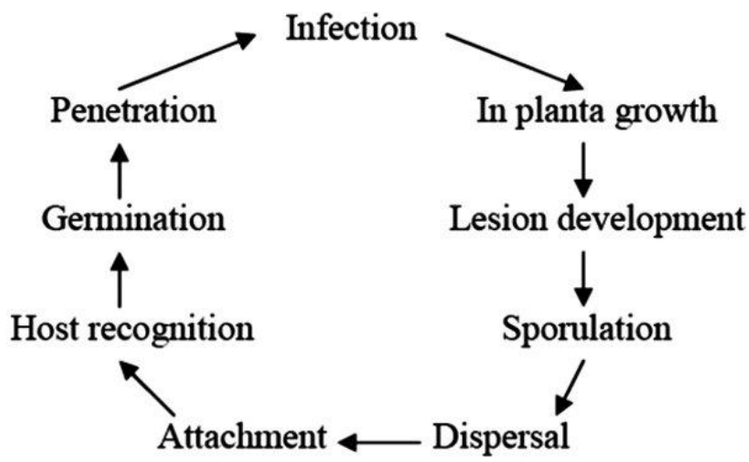
ROLE OF POLYSACCHARIDES IN PATHOGENESIS

Polysaccharides: Fungi, bacteria and nematodes release varying amounts of

mucilaginous substances that coat their bodies and provide interface between the outer surface of the micro-organism and its environment. The role of slimy polysaccharides is of utmost importance in wilt diseases. In the vascular wilts, large polysaccharide molecules released by the pathogen in the xylem causes mechanical blockage of vascular bundles and initiate wilting.
Ex: *Ralstonia solanacearum* (Bacterial wilt of Solanaceous plants)

1.3. DISEASE CYCLE

The disease cycle is a series of interconnected events, which lead to the disease development and pathogen propagation. These events include inoculation, prepenetration, penetration, infection, colonization (invasion), and growth and reproduction of the pathogen



The stages involved in disease cycle are outline below.

i. Inoculation

This is thetransfer of the pathogen inoculum to the infection court, or area in which invasion of the plant occurs (the infection court may be the unbroken plant surface, a variety of wounds, or natural openings—e.g., stomata [microscopic pores in leaf surfaces], hydathodes [stomata-like openings that secrete water], or lenticels [small openings in tree bark]

In some cases, the inoculum is represented as an intact cell as in bacteria, mollicutes, protozoa, viruses, and viroids. There are two types of inoculum: **primary** and **secondary inoculum**, which in turn cause primary and secondary infection. The **primary inoculum** lives dormant in the winter or summer and causes the original infections in the spring or in the autumn.

The **secondary inoculum** is that one which is produced from primary infections. The primary inoculum is more abundant than secondary inoculum and closer to the crop, and caused more severe diseases and the losses that result.

ii. Pre-penetration

Attachment of pathogen to host;

Some pathogens directly penetrate the plant tissues by their vectors and then are surrounded by cytoplasm, cell membrane, or cell wall of plant cell, such as mollicutes, fastidious bacteria, protozoa, and most viruses. In other cases, the pathogen firstly makes contact with the external surface of the plant, and then penetration process takes place, such as fungi, bacteria, and parasitic higher plants. The adhesion of the pathogen with plant surface is carried out by mucilaginous substances found on the pathogen surface or at its tip. These substances are composed of mixture of water-insoluble

polysaccharides, glycoproteins, and lipids, which, when moistened, become sticky and help the pathogen adhere to the plant.

iii. Penetration

Phytopathogens penetrate plant surfaces either through natural openings such as fungi and nematodes or through wounds in cell wall such as bacteria, viruses, viroids, mollicutes, fastidious bacteria, and protozoa. Penetration and infection are not usually correlated together, because some penetrated plants are resistant to phytopathogens [26].

iv. Spore germination

Spore germination process initiates by growth stimulation, which takes place with the availability of proper environmental conditions. Once the stimulation has been received, the spore starts to utilize the stored food, such as lipids, and carbohydrates to build germ tube as abridge with cell membrane and cell wall of the plant. When appropriate physical and chemical signals, such as surface hardness, surface topography, and plant signals, are present, germ tube extension and differentiation take place

v. Recognition between host and pathogen

When a pathogen comes in contact with a host cell, the plant triggers a signal that either allows or retards the pathogen growth and development of disease. This signal is a biochemical reaction, which acts as a receptor to a pathogen contact. The pathogen propagation depends on the components of the plant cell, such as fatty acids galacturonan, phenolic compounds, amino acids, and sugars .

vi. Spores and seed germination

The availability and ability of host infection are increased by vegetative pathogen. The infection by fungal spores or parasitic higher plant seeds is carried out after germination has achieved. Fungal spores' germination is carried out by releasing either a mycelium or a germ tube that grows into the plant cell and cause infection.

vii. Growth of nematodes

The growth of nematodes starts with hatching of eggs, which essentially requires convenient environmental conditions such as temperature and moisture. After hatching of the eggs, the larvae penetrate the plant cell and grow to form the adults. After maturation, the adults of nematodes closely adhere with plant roots due to some factors, such as carbon dioxide and amino acids

viii. Infection

Infection is the process by which pathogens establish contact with susceptible cells or tissues of the host and procure nutrients from them. The intimate contact of phytopathogen with its host is called **infection process**. The infection process is either successful or unsuccessful depending on the type of host, whether susceptible or resistant, respectively. Successful infection results in the appearance of symptoms, such as discoloration, necrosis, dwarfism, and so on of the host.

While unsuccessful (latent) infection does not lead to any observations for the symptoms. As well known, the symptoms start to appear after the incubation period of the pathogen has been finished. The symptoms either are stable or may be changed until death .

1.3.1 Growth and reproduction of the pathogen

Most phytopathogens especially fungi and parasitic higher plants invade and infect plant tissues through the point of inoculation. Therefore, these pathogens can easily grow and spread within the plant tissues until a certain limit or death occurs. For example, fungi can invade and infect the plant tissue to cause vascular wilts. This invasion is carried out by releasing spores within the vessels.

The time interval between inoculation and the appearance of disease symptoms is called the **incubation period**.

1.4 PARASITISM AND PATHOGENICITY

A parasite is an organism that lives on or in some other organism and obtains its food from there. The removal of food by a parasite from its host is called **parasitism**. Therefore, a plant parasite is an organism that becomes intimately associated with a plant and multiplies or grows at the expense of the plant.

Parasites removes water and nutrients from the host plant and reduces efficiency in the normal growth of the plant and becomes detrimental to the further development and reproduction of the plant.

In certain cases both the plant and the microorganism benefit from the association or relationship and this phenomenon is known as **symbiosis.**These parasites are successful because they can invade a host plant, feed and proliferate in it, and withstand the conditions in which the host lives.

Some parasites, including viruses, viroids, bacteria, nematodes, protozoa,and fungi that causes various diseases in plants like downy mildews, powdery mildews, and rusts, are called **biotrophs** (they can only live and multiply on another living organism)and are referred to as **obligate parasites**

Other parasites (most fungi and bacteria) can live on either living or dead hosts and on various nutrient media, and they are therefore called **non-obligate parasites**.

Some parasites spend most of the time and thrive well on dead organic matter (**necrotrophs)** but, under certain circumstances, may attack living plants and become parasitic; these parasites are called **facultative parasites.**

1.5 DISSEMINATION OF A PATHOGEN

A few pathogens, such as nematodes, fungi, and bacteria, can move short distances using their own power and thus can move from one host to another one very close to it.

However, there are several ways that contributes to the spread of pathogens from one area to another or form an infected plant to a health plant. These are:

i. By Vectors of Insects, Mites, Nematodes, Birds and Other Vectors

Insects, particularly aphids, leafhoppers, and whiteflies, are by far the most important vectors of viruses, whereas leafhoppers are the main vectors of bacteria, and protozoa. Each one of these pathogens is transmitted, internally, by only one or a few species of insects during feeding and movement of the insect vectors from plant to plant.

The insects carry these pathogens externally from plant to plant and deposit them on the plant surface or in the wounds they make on the plants during feeding.

ii. Dissemination by Humans

Human beings disseminate all kinds of pathogens over short and long distances in a variety of ways. Within a field, humans disseminate some pathogens, such as tobacco mosaic virus, through the successive handling of diseased and healthy plants. Inaddition, human beings may also carry pathogens on the clothing and working shoes and introduce in another area.

iii. Dissemination by air

Spores of most oomycetes (a subclass of parasitic fungi that includes water molds, white rusts, and downy mildews) and most fungi and the seeds of most parasitic plants are disseminated by air currents that carry them as inert particles over long distances.

iv. Dissemination by water

Water is important in disseminating pathogens in three ways. Firstly, Bacteria, nematodes, and spores of fungi present in the soil are disseminated by rain or irrigation water that moves on the surface or through the soil. Secondly,all bacteria and the spores are spread by the rain or (overhead) irrigation water, which either washes them downward or splashes them in all directions.

Thirdly, raindrops or drops from overhead irrigation pick up the fungal spores and any bacteria present in the air and wash them downward, where some of them may land on susceptible plants.

v. Dissemination by farming implements

Pathogens can also be disseminated through tools, such as contaminated pruning equipment used on diseased plants and then carried to healthy plants. Farming equipments such as tractors and hoes can aid the transportation of pathogens from infected arrears to healthy areas.

vi. Using infected planting materials

Infected transplants, cuttings, suckers, seed, and nursery stock when transported from one region to another can disseminate pathogens.

LECTURE 4.

1.0 DEFENCE MECHANISM IN PLANTS AGAINST PATHOGENS

It is very important to understand that whatever kind of defense or resistance a host plant employs against a pathogen or against an abiotic agent, it is ultimately controlled, directly or indirectly, by the genetic material (genes) of the host plant and that of the pathogen. Therefore, the following are some of the mechanisms plants use to defend themselves.

i. Non-host Resistance

A plant may find it easy to defend itself against pathogens when it is brought in contact with a pathogenic biotic agent to which the plant is not a host.This is known as non-host resistance and is the most common form of resistance. For example, apple trees are not affected by pathogens of potatoes, tomato, maize and so on because the genetic makeup of apple is in some way(s) different from that of host plants, which, of course, they are attacked by their own pathogens.

ii. Preexisting Defense Structures

The first line of defense of a plant against pathogens is its surface, which the pathogen must adhere to and penetrate if it is to cause infection. Some structural defenses are present in the plant even before the pathogen comes in contact with the plant. Some of them are

-**structure and thickness of the epidermal cell walls**(thick walls makes direct penetration by fungal pathogens difficult or impossible

-size, location, and shapes of stomata (if more stomata are located on the underside of the leaf then few vectors will have access, also varieties that have stomata that do not open much or open for a shorter period offer some form of defense as the pathogen cannot penetrate)

--**quality of wax** (forms a water repellent surface that prevent the pathogen to be deposited)

- thick cuticle (increase resistance to infection in diseases in which the pathogen enters its host only through direct penetration)

-presence of tissues made of thick-walled cells that hinder the advance of the pathogen on the plant

- Hairy leaf and stem surface (prevents the vectors to land there and reduce and prevent the pathogens to be deposited)

-presence of thorns (this also deters the vector to land on such plants)

iii. Preexisting Chemical Defenses

A particular pathogen will not infect certain plant varieties even though no structural barriers of any kind seem to be present or to form in these varieties. This clearly suggest that defense mechanisms of a chemical rather than a structural nature are responsible for the resistance to infection exhibited by plants against certain pathogens.

- **Inhibitors Released by the Plant in Its Environment:** Certain plants release a variety of substances through the surface of their aboveground parts as well as through the surface of their roots. Some of the compounds released by certain kinds of plants, however, seem to have an inhibitory action against certain pathogens. Fungitoxic exudates on the leaves of some plants, e.g., tomato and sugar beet, seem to be present in sufficient concentrations to inhibit the germination of spores of fungi

- **Inhibitors Present in Plant Cells before Infection:** some plants are resistant to diseases caused by certain pathogens because of one or more inhibitory antimicrobial compounds, known as ***phytoanticipins***, which are present in the cell before infection.

iv. Defense through lack of essential factors

-Lack of Recognition between Host and Pathogen: Plants of a species or variety may not become infected by a pathogen if their surface cells lack specific recognition factors (specific molecules or structures) that can be recognized by the pathogen. If the pathogen does not recognize the plant as one of its host plants, it may not become attached to the plant or may not produce infection substances.

-Lack of Host Receptors and Sensitive Sites for Toxins: thetoxin, which is responsible for the symptoms, is thoughtto attach to and react with specific receptors or sensitive sites in the cell. Only plants that have such sensitivereceptors or sites become diseased. Plants of other varieties or species that lack such receptors or sites remainresistant to the toxin and will develop no symptoms.

- Lack of Essential Substances for the Pathogen: plants that for some reason do not produce one of the substances essential for the survivalof an obligate parasite would be resistant to the pathogen thatrequires that subatance for survival.

v. Induced structural defenses

Plants usually respond by forming one or more types of structures that are more or less successful in defending the plant from further pathogen invasion even after penetration of the pathogen through wounds and natural openings like stomata.

-Some of the defense structures formed involve the cytoplasm of the cells under attack where the cytoplasm becomes dense and disintegrates the pathogen, and the process is called ***cytoplasmic defense reaction***;

-some defense involve the walls of invaded cells which undergoes morphological changes and are called ***cell wall defense structures***.

-Further, death of the invaded cell may protect the plant from further invasion. This is called the ***necrotic*** or ***hypersensitive defense reaction.***

-Formation of Abscission Layers: this is the layer formed on young, active leaves of stone fruit trees after infection by any of several fungi, bacteria, or viruses. The plant will discard the infected area along with a few yet uninfected cells, protects the rest of the leaf tissue from being invaded.

-Deposition of the gums**:** The defensive role of gums stems is that they are deposited quickly in the intercellular spaces and within the cells surrounding the region of infection, thus forming an impenetrable barrier that completely encloses the pathogen. The pathogen then becomes isolated, starves, and sooner or later dies.

-Abscission layers; An abscission layer consists of a gap formed between infected and healthy cells of a leaf surrounding the locus of infection due to the disintegration of the middle lamella of parenchymatous tissue.

vi. Genetically engineering disease-resistant plants

The number of plant genes for resistance (R genes) that have been isolated is increasing rapidly across the world. Scientists have managed to isolate and make certain plant species more resistant to certain pathogens.

2. CULTURAL PRACTICES AND CONTROL MEASURES AIM AT REDUCING PLANT DISEASES

i. Site Selection and Preparation

Low-lying and poorly drained and aerated fields, especially if near other infected fields, tend to favor the appearance and development of epidemics.

ii. Selection of Propagative Material

The use of seed, nursery stock, and other propagative material that carries various pathogens increases the amount of initial inoculum within the crop and favors the development of epidemics greatly.

The use of pathogen-free or treated propagative material therefore, can reduce the chance of epidemics greatly.

iii. Cultural Practices

Continuous monoculture, large hectors planted with the same variety of crop, high levels of nitrogen fertilization, no-till culture, dense plantings, overhead irrigation, injury by herbicide application, and poor sanitation all increase the possibility and severity of epidemics because the conditions become very favorable for disease development.

iv. Disease Control Measures

Chemical sprays, cultural practices (such as sanitation and crop rotation), biological controls (such as using resistant varieties), and other control measures reduce or eliminate the possibility of a disease.

v. Introduction of new Pathogens

The frequency of worldwide travel have also increased the movement of seeds, tubers, nursery stock, and other agricultural goods. These activities increase the possibility of introducing pathogens into areas where the hosts have not had a chance to evolve resistance to these pathogens. Such pathogens frequently lead to severe epidemics.

vi. Inspection and quarantine

Rigorous inspection measures have been put in place by authorities to ensure that only seeds and plant products that meet standards are allowed to reach the market or entry into the country or region.

vii. Crop Certification

Several voluntary and compulsory inspection systems are in effect in various countries to inspect and issue certificates for seed and plant products. Only those which are certified and tested reach the market and stiff penalties are in place to those who evade the inspection and certification process.

LECTURE 5

1.0 COMMON PLANT DISEASES AND THEIR SYMPTOMS

1.1 FUNGAL DISEASES

1.1.1 Introduction

Fungi are eukaryotic organisms; i.e., their cells contain membrane-bound organelles and clearly defined nuclei. They are small, generally microscopic and usually filamentous, branched, spore-bearing organ- isms that lack chlorophyll. Fungi have cell walls that contain chitin and glucans (but no cellulose) as the skeletal components.

Some fungi, known as **obligate parasites or biotrophs**, can grow and multiply only by remaining, in association with their host plants in their entire life. Others, known as **non-obligate parasites,** require a host plant for part of their life cycles but can complete their cycles on dead organic matter, or they can grow and multiply on dead organic matter as well as on living plants.

1.1.2 Characteristics of plant pathogenic fungi

i. Morphology

Most fungi have a filamentous vegetative body called a **mycelium.** The mycelium branches out in all directions. The individual branches of the mycelium are **called hyphae.**

ii. Reproduction

Fungi reproduce chiefly by means of spores. Spores are reproductive bodies consisting of one or a few cells and may be formed asexually,or as the result of sexual fertilization.

In some fungi, **asexual spores** are produced inside a sac called a ***sporangium***. Some of these spores can swim by means of flagella and are called ***zoospores***. Other fungi produce asexual spores called **conidia** by the cutting off of terminal or lateral cells from special hyphae called **conidiophores.**

Sexual reproduction occurs in most groups of fungi. In Zygomycetes, two cells (gametes) of similar size and appearance unite and produce a zygote called a **zygospore.**

iii. Dissemination

Zoospores are the only fungus structures that can move by themselves. Zoospores, however, can move for only very short distances like few millimeters or centimeters in rare instances. Only **myxomycetes, oomycetes**, and **chytridiomycetes** produce zoospores. The great majority of plant pathogenic fungi depend for their spread from plant to plant and to different parts of the same plant on hyphal growth or chance distribution by such agents as wind, water, birds, insects, other animals, and humans. Fungi are disseminated primarily in the form of spores.

1.1.3 Signs and symptoms caused by fungi on plants

In most cases, Fungi cause local or general symptoms on their hosts. As such symptoms may occur separately or concurrently or may follow one another.

In general, fungi cause local or general **necrosis** of plant tissues, and they often cause reduced growth **(stunted growth or hypotrophy)** of plant organs or entire plants. A few fungi cause excessive growth of infected plants or plant parts (***hypertrophy or hyperplasia***) .

Symptoms associated with necrosis include *leaf spots, blight, scab, rots, damping-off, dieback, and canker*. Symptoms associated with hyperplasia include galls, warts, and leaf curls.

In some instances, the fungus infecting the plant may produce growth or structures on the plant, stems, or leaves such as masses of mycelium or aggregates of spores with a characteristic appearance. These developments are referred to as **signs of infection**, in contrast to symptoms, which refer specifically to **the plant or plant tissue.**

The most commonnecrotic symptoms of fungal infection are as follows:

i. **Leaf spots**: Localized lesions on host leaves consisting of dead and collapsed cells
ii. **Blight**: this causes general and extremely rapid browning and death of leaves, branches, twigs, and floral organs.
iii. **Canker**: Localized necrotic lesion on a stem or fleshy organ, often sunken, of a plant
iv. **Dieback**: Extensive necrosis of twigs beginning at their tips and advancing toward their bases
v. **Root rot:** Disintegration or decay of part or all of the root system of a plant
vi. **Scab:** Localized lesions on host fruit, leaves, tubers, etc., usually slightly raised or sunken and cracked, giving a scabby appearance
vii. **Soft rots and dry rots**: Maceration and disintegration of fruits, roots, bulbs, tubers, and fleshy leaves
viii. **Damping-off**: Rapid death and collapse of very young seedlings
ix. **Basal stem rot**: Disintegration of the lower part of the stem
x. **Galls**: Enlarged portions of plant organs (stems, leaves, blossoms, roots)
xi. **Warts**: Wart-like protuberances on tubers and stems
xii. **Witches'-brooms**: Profuse, upward branching of twigs
xiii. **Leaf curls**: Distortion, thickening, and curling of leaves
xiv. **Wilt:** Generalized loss of turgidity and drooping of leaves or shoots
xv. **Rust:** Many small lesions on leaves or stems, usually of a rusty color
xvi. **Smut:** Seed or a gall filled with the mycelium or black spores of the smut fungi
xvii. **Mildew**: Areas on leaves, stems, blossoms, and fruits, covered with whitish mycelium

In many diseases, the fungal pathogen grows, or produces various structures on the surface of the host. These structures may include mycelium, fruiting bodies, and spores, and are called signs. Signs refer only to the appearance of the infected plants or plant tissues.

For example in mildews, one sees mostly the signs consisting of a whitish, downy or powdery growth of fungus mycelium and spores on the plant leaves, fruit, or stem, whereas the symptoms consist of chlorotic or necrotic lesions on leaves, fruit, and stem, reduced growth of the plant, and so on.

To be sure that the disease is fungal, part of the infected parts e.g leaf or fruit can be isolated and put on the petri-dish for laboratory analysis using scientific methods such as **Polymerise Chain Reaction (PCR).**



Witches broom





Late blight in tomatoes



Fruit cankers



Powdery mildew on tomato



Damping off

1.2 BACTERIAL DISEASES

1.2.1 Introduction

Bacteria are generally single-celled microorganisms whose genetic material (DNA) is not bound by a membrane and therefore is not organized into a nucleus.

Most bacteria are strictly saprophytic and as such are beneficial to humans because they help decompose the enormous quantities of organic matter produced yearly by humans, animals, and factories as waste products or by the death of plants and animals. Several species cause diseases in humans, such as tuberculosis, pneumonia, and typhoid fever, and a similar number cause diseases in animals, such as brucellosis and anthrax.

Bacteria are in different shapes, some are rod shaped, spherical, spiral, or filamentous (threadlike). Some bacteria can move through liquid media by means of ***flagella***, whereas others have no flagella and cannot move themselves

1.2.2 Characteristics of plant pathogenic bacteria

i. Morphology

Majority of the plant pathogenic bacteria are rod-shaped.

The cell walls of bacteria of most species are enveloped by a viscous layer forming a definitive mass around the cell, is called a ***capsule.*** All the material inside the cell wall constitutes the protoplast.

Rod-shaped phyto-pathogenic bacteria reproduce by the asexual process known as ***binary fission, or fission.***

This occurs by the inward growth of the cytoplasmic membrane toward the center of the cell, forming a transverse membranous partition dividing the cytoplasm into two approximately equal parts. The two layers of cell wall material, are then synthesized between the two layers of the membrane.

When the formation of these cell walls is completed, the two layers separate, splitting the two cells apart.

Bacteria reproduce at an **astonishingly rapid rate**. Under favorable conditions, bacteria may divide every 30 to 50 minutes, one bacterium becoming 2, 2 becoming 4, 4 becoming 8, 8 become 16 and so on. Over a million bacteria can be produced in a day.

ii. Transmission and infection

In order for a bacterium to produce a disease in a plant, the bacterium must first invade the plant tissue and multiply. Bacterial pathogens enter plants through wounds, principally produced by adverse weather conditions, humans, tools and machinery, insects, and nematodes, or through natural openings such as stomata, lenticels (pores in the stem for gaseous exchange), hydathodes, nectar-producing glands, and leaf scars.

Most foliage invaders are spread from plant to plant by windblown rain or dust. Humans disseminate bacteria through *cultivation, grafting, pruning, and transporting diseased plant material*. Animals, including insects and mites, are other common transmission agents

When conditions are unfavorable for growth and multiplication, bacteria remain dormant on or inside plant tissue. Some, such as the **crown gall bacterium**, may survive for months or years in the soil.

Bacterial diseases are influenced greatly by **temperature and moisture**. Often, a difference of only a few degrees in temperature determines whether a bacterial disease will develop. In most cases, moisture as a water film on plant surfaces is essential for establishing an infection.

iii. Ecology and spread

Almost all plant pathogenic bacteria develop mostly in the host plant as parasites, on the plant surface, especially buds, as **epiphytes**, and partly in plant debris or in the soil as **saprophytes.**

The main classes of bacteria that causes diseases in plants are **agrobacteria** which are rod-shaped and possess flagella, **clavibacter** which are mainly non-motile, ***erwinia*** which are straight rods and are motile by means of flagella and the last group is **pseudomonas** which are straight or curve shaped and are most abundant in soils, fresh water and marine environments. Other are ralstonia, xylella and xanthomonas groups.

1.3 Symptoms and signs caused by bacterial disease

Bacterial diseases can be grouped into **four** broad categories based on the extent of damage to plant tissue and the symptoms that they cause, which may include:

(i) Vascular wilt,

Vascular wilt results from the bacterial invasion of the plant's vascular system. The subsequent multiplication and blockage prevents movement (translocation) of water and nutrients through the xylem of the host plant. Drooping, wilting, or death of the aerial plant structure may occur; examples include bacterial wilt of sweet corn, tobacco, tomato, and cucurbits (e.g., squash, pumpkin, and cucumber).

(ii) Necrosis

Pathogens can cause necrosis by secreting a toxin (poison). Symptoms include formation of leaf spots, stem blights, or cankers.

(iii) Soft rot

Soft rot diseases are caused by pathogens that secrete enzymes capable of decomposing cell wall structures, thereby destroying the texture of plant tissue—i.e., the plant tissue becomes macerated (soft and watery). Soft rots commonly occur on fleshy vegetables such as potato, carrot, eggplant, squash, and tomato.

(iv)Tumors.

Tumor diseases are caused by bacteria that stimulate uncontrolled multiplication of plant cells, resulting in the formation of abnormally large structures.

1.3Examples of some specific bacterial diseases

Below is a list of examples of some common bacterial diseases.

i. Bacterial spots and blight

These appearas spots of various sizes on leaves, stems, blossoms, and fruits. In some bacterial diseases the spots continue to advance rapidly and the diseases are then called blights.

In severe infections the spots may be so numerous that they destroy most of the plant surface and the plant appears blighted, thus producing large areas of dead plant tissue and blighted plants.

In **dicotyledonous plants** the bacterial spots on some hosts are restricted by large veins, and the spots appear angular. For the same reason, bacterial spots on monocotyledonous plants appear as streaks or stripes.



Bacterial leaf spot on pepper



Bacterial spot (xanthomonas)



Tomato bacterial blight



Bacterial blight in beans



Galls on fruit trees



Crown Gall on fruit tree (*Agrobacterium tumefaciens*)

1.4 Management and control of plant diseases

Bacterial diseases of plants are usually very difficult to control. Frequently, a combination of control measures is required to combat a given bacterial disease. The following measures can help to control bacterial as well as other plant diseases.

- **Chemical controls**: involve the use of chemicals like fungicides, bactericides to control plant disease. Chemical controls can be effective, but it is important to properly identify the problem which may require submitting samples to a diagnostic lab for testing.
- **Genetic controls**: involve breeding disease resistant plants which creates new cultivars or varieties that perform better.
- **Cultural practices**: this involves the use of various practices that aim at avoiding the provision of conducive environment for the survival and multiplication of the pathogen. For example, crop rotation to disrupt year-to-year pest cycles, sanitizing tools after using them on diseased plants to prevent further spread, planting warm seasoned crops after the soil has warmed to prevent root rot, watering plants in the morning so they can dry during the day and remain dry through the night to prevent fungal, and bacterial diseases, and pruning and trimming plants in such a way that it promotes airflow and more light penetration to prevent infection.
- **Regulatory controls:**these are based on exclusion of pathogenic organisms, highly susceptible host plants, and alternate host plants from a particular area.
- **Biological controls**: this essentially involve using nature to prevent or treat diseases
- **Sanitation practices:** aiming at reducing the inoculum in a field by removing and burning infected plants or branches, and at reducing the spread of bacteria from plant to plant. This includes cleaning of the farm equipment and proper weeding.
- **Adjusting fertilizing and watering**: this is done so that the plants are not extremely succulent during the period of infection may also reduce the incidence of disease
- **Crop rotation**: this can be very effective with bacteria that have a limited host range, but is impractical and ineffective with bacteria that can attack many types of crop plants
- **Seed treatment** with hot water at about 50 °C (120 °F) is also effective against management of bacterial diseases in carrots, eggplant, pepper and tomatoes.
- **destruction** of all plant materials that may harbor pathogenic fungi
- **Biosecurity measure** are also important to avoid the transfer of disease pathogens from an infected area to another.
- **Isolation** or, as it is usually called, purification of viruses is a very effective method of managing viral pathogens.

LECTURE 6.

1.0 DISEASES CAUSED BY VIRUSES

1.1 Introduction

A virus is a nucleoprotein that multiplies **only** in living cells and has the ability to cause disease. It is too small to be seen individually with a light microscope. All viruses parasitize cells and cause a multitude of diseases in all forms of living organisms.

Viruses consist of nucleic acid and protein, with the protein forming a protective coat around the nucleic acid. Although viruses can take any of several forms, they are mostly **rod shaped**.

1.2 Characteristics of plant viruses

From the onset, it is important to note that plant viruses differ greatly from all other plant pathogens not only in size and shape, but also in the simplicity of their chemical constitution and physical structure, methods of infection, multiplication, translocation within the host, dissemination, and the symptoms they produce on the host. Because of their small size and the fact that they are transparent, viruses generally cannot be viewed and detected by the methods used for other pathogens.

Viruses are not cells nor do they consist of cells.

1.3 Detection

Virus are very difficult to detect and to see under light microscope.

The present methods of detecting plant viruses involve primarily the transmission of the virus from a diseased to a healthy plant by budding or grafting, or by rubbing leaves of healthy plants with sap from an infected plant. Certain other methods of transmission, such as insect vectors, are also used to demonstrate the presence of a virus. Most of these methods, however, cannot distinguish whether the pathogen is a virus or bacterium.

Therefore, the most definitive proof of the presence of a virus in a plant is provided by purification, **electron microscopy**, and, most commonly, **serology**.

1.4 Morphology

Plant viruses come in different shapes and sizes. Nearly half of them are elongate (rigid rods or threads), and almost as many are spherical (isometric).Each plant virus consists of at least a nucleic acid and a protein.

1.5Transmission and infection

Plant viruses enter cells only through **wounds** made mechanically or by vectors or by deposition into an ovule by an infected pollen grain.

When a virus infects a plant, it moves from one cell to another and multiplies in most, if not all of such cells.

Viruses move from cell to cell through the**plasmodesmata**(*small channels that directly connect the cytoplasm of neighboring plant cells to each other, and therefore establishes living bridges between cells*)connecting adjacent cells. Viruses multiply in each parenchyma cell they infect.

In all economically important viral infections, viruses reach the phloem and through it are transported rapidly over long distances within the plant. Most viruses, however, require **2 to 5 days or more** to move out of an inoculated leaf.

Once the virus has entered the phloem, it moves rapidly in it toward growing regions (apical meristems) or other food-utilizing parts of the plant, such as tubers and rhizome. It later spreads throughout the plant.

1.6 Symptoms caused by plant Viruses.

Plant viruses are transmitted from plant to plant in a number of ways. These are;-

Modes of transmission include vegetative propagation, mechanically through sap, through seed, pollen, and by specific insects, mites, nematodes, and fungi.

Almost all viral diseases seem to cause some degree of dwarfing or stunting of the entire plant and reduction in total yield.

Viruses usually shorten the length of life of virus-infected plants, although they rarely kill plants on infection.

The most obvious symptoms of virus-infected plants usually appear on the leaves.

In almost all virus diseases of plants occurring in the field, the virus is present throughout the plant *(systemic infection*) and induces the formation of systemic symptoms.

It is important to understand that many viruses infect certain hosts without causing development of visible symptoms on them. Such viruses are usually called **latent viruses**, and the hosts are called **symptomless carriers.**

1.7 Classification of plant viruses

Many plant viruses are named after the most conspicuous symptom they cause on the first host in which they have been studied. Thus, a virus causing a mosaic on tobacco is called **tobacco mosaicvirus**, whereas the disease itself is called tobacco mosaic; another virus causing spotted wilt symptoms on tomato is called **tomato spotted wilt virus** and the disease is called tomato spotted wilt, and so on.

All viruses belong to the kingdom **Viruses**. Within the kingdom, viruses are distinguished as **RNA** viruses and **DNA** viruses, depending on whether the nucleic acid of the virus is **RNA or DNA**. Viruses are further subdivided depending on whether they possess one or two strands of RNA or DNA of either positive or negative sense, either filamentous or isometric.

1.8 Control of plant viruses

- The best way to control a virus disease is by keeping it out of an area through systems of quarantine, inspection, and certification.
- Eradication of diseased plants to eliminate inoculum from the field may, in some cases, help control the disease.
- Plants may be protected against certain viruses by protecting them against the virus vectors.
- The use of virus-free seed, tubers, cuttings, and so on is the single most important measure for avoiding virus diseases of many crops, especially those lacking insect vectors.
- Breeding plants for hereditary resistance to viruses is of great importance, and many plant varieties resistant to certain virus diseases have already been produced.
- Some viruses can be inactivated by heat. Dormant, propagative organs are usually dipped in hot water (35–54°C) for a few minutes or hours, whereas actively growing plants are usually kept in greenhouses or growth chambers at 35 to 40°C for several days, weeks, or months, after which the virus in some of them is inactivated and the plants are completely healthy.

- No chemical substances (viricides) are yet available for controlling virus diseases of plants in the field, although some compounds can help reduce the problem.

1.9 Examples of Diseases caused by viruses

i. Tobacco mosaic virus

Symptoms consist of various degrees of mottling, chlorosis, curling, distortion, and dwarfing of leaves, flowers, and entire plants.

On tomato, leaflets may become long and pointed and, sometimes, shoestring like. Infections of young plants reduce fruit set.



Tobacco mosaic virus

ii. Bean Common Mosaic and Bean Yellow Mosaic

Both bean common mosaic and bean yellow mosaic occur wherever beans is grown. Bean common mosaic affects all beans, but only beans (*Phaseolus vulgaris and some other Phaseolus spp*.), whereas bean yellow mosaic also affects peas.

They are often found in the same field and often on the same plants. Plants infected with either virus may show mottling, yellowing, and malformation of leaves and pods.

Yields may be reduced by up to 80 to 100%, depending on the plant stage of growth at the time of infection.



Bean common mosaic virus

iii. Potato virus Y (PVY)

Potato virus Y (PVY) occurs worldwide and is of great economic importance. It affects potato, pepper, tomato, and tobacco and causes severe losses on all these hosts. Symptoms vary from a mild to severe mottle on most hosts to a streak or “leafdrop streak” resulting from long necrotic lesions along the veins on the underside of leaflets of some potato varieties.

It is transmitted mainly through infected potato seed tubers.



Potato virus Y

iv. Potato Leafroll

Potato leafroll occurs worldwide. It is caused by the potato leafroll virus (PLRV) and affects only potato. It causes high yield losses and can be the most devastating virus of potato. It causes a prominent upward rolling of the leaves, and the plants are stunted and have a stiff upright growth.

The virus is transmitted through infected potato seed tubers and, in the field, by more than 10 species of aphids in the persistent manner. Its control depends on the use of potato seed tubers free of the virus. Because the vector must feed for several hours to acquire the virus and for several more hours to infect the plant with the virus, some control of PLRV has been achieved through early control of the aphid vectors with insecticides.



v. Maize Streak Virus (MSV)

Maize streak is widespread and severe in the southern half of Africa.

It appears as spots on the leaves that later develop into streaks. Plants infected are stunted and their ears are poorly filled. The maize streak geminivirus is transmitted by several species of **leafhoppers**.



Maize streak virus

vi. Cassava mosaic virus (CMV)

This disease is present in all regions where cassava is grown and causes huge losses.

Cassava plants infected with CMV produce leaves that exhibit mild to severe mosaic symptoms

If plants are infected young they tend to show more severe mosaic, their leaf blades are narrow, the entire plants remain stunted and they form small tubers that contain only a small amount of starch. In older plants, symptoms are milder or absent.

It is transmitted in nature through infected cassava stem cuttings used as propagative material and by the whitefly called *Bemisiatabaci*.



Cassava mosaic virus

2.0 DISEASES CAUSED BY VIROIDS

Viroids are small, low molecular weight ribonucleic acids (RNAs) that can infect plant cells, replicate themselves, and cause disease.

Viroids differ from viruses in at least two main **characteristics:**

(i) the size of RNA in viroids are much smaller compared to viruses
(ii) the fact that virus RNA is enclosed in a protein coat whereas viroids lack a protein coat and apparently exist as free (naked) RNA.

How viroids replicate themselves is still not known and howviroids cause disease is also not known. However, some research evidence shows that viroids replicate by direct RNA copying in which all components required for viroid replication, including the RNA polymerase, are provided by the host.

Viroid diseases show a variety of symptoms that resemble those caused by virus infections.

2.1.1 Transmission

Viroids are spread from diseased to healthy plants primarily by mechanical means, i.e., through sap carried on **hands or tools during propagation** or cultural practices and, of course, by **vegetative propagation.** Some, such as potato spindle tuber, chrysanthemum stunt, and chrysanthemum chlorotic mottle viroids, are transmitted through **sap** quite readily, whereas others, such as citrus excortis viroid, are transmitted through sap with some difficulty.

Several viroids, e.g., those causing potato spindle tuber, cadang-cadang, tomato bunchy top, and apple scar skin, appear to be transmitted through **the pollen and seed**, but the rates of such transmission are usually very small.

Viroids also seem to be transmitted on the **mouthparts or feet of some insects.**

Viroids apparently survive in nature outside the host or in dead plant matter for periods of time varying from a few minutes to a few months.

2.1.2 Control of viroids

The control of diseases caused by viroids is based on the use of viroid-free propagating stock, removal and destruction of viroid-infected plants, and washing of hands or sterilizing of tools after handling viroid infected plants before moving on to healthy plants.

2.1.3 Examples of viroid plant diseases

i. Potato spindle tuber viroid (PSTVd)

Potato spindle tuber viroid (PSTVd) is the first recognized viroid.

It causes severe losses, and in some regions it is one of the most destructive diseases of potatoes. It attacks all varieties and spreads rapidly.

Infected potato plants appear erect, and dwarfed

The leaves are small and erect, and the leaflets are darker green and sometimes show rolling and twisting.

The tubers are elongated, with tapering ends

Tubers are smoother, but tuber eyes are more numerous and more conspicuous.



Potatoe spindle tuber

The viroid is mechanically transmissible and is spread primarily by knives used to cut healthy and infected potato seed tubers and during handling and planting of the crop.

Potato spindle tuber can be controlled effectively by planting only PSTVd-free potato tubers in fields free of diseased tubers that may have survived from the previous year's crop.

3.0 DISEASES CAUSED BY NEMATODES

3.1 Introduction

Nematodes belong to the kingdom Animalia. Nematodes are wormlike in appearance but quite distinct taxonomically from the true worms. Most of the several thousand species of nematodes live freely in fresh or salt waters or in the soil, and feed on microorganisms and microscopic plants and animals. They are extremely small and as such can be seen with the naked eye but they can be observed under a microscope. The nematode body is more or less transparent.

Several species of nematodes, are known to feed on **living plants**, obtaining their food with spears or stylets and causing a variety of plant diseases worldwide.

Ecological classification of nematodes
There are two major classes
I. Above ground feeder
II. Below ground feeder

I. Above ground feeder
a. Feeding on flower buds, leaves and bulbs
i) Seed gall nematode:
Anguina tritici
ii) Leaf and bud nematode:
Aphelenchoides
iii) Stem and bulb nematode:
Dictylenchus

b. Feeding on tree trunk
i) Red ring nematode:
Rhadinaphelenchus cocophilus
ii) Pine wilt nematode:
Bursaphelenchus xylophilus

II. Bellow ground feeder
It is again classified in to three classes
I) Endoparasitic nematodes
II) Semiendoparasitic nematodes
III) Ectoparasitic nematodes

a) Endoparasitic nematodes
The entire nematode is found inside the root and the major portion of nematode body found inside the plant tissues. They are two types
1) Migratory endoparasite
:These nematodes move in cortical parenchyma of host root. While migrating they feed on cells, multiply and cause necrotic lesion. Example,
Pratylenchus spp.,
Radopholus spp. and
Hirschmanniella spp.

2) Sedentory endoparasite
: the second stage larvae penetrate the root lets and become sedentary throughout the life cycle, inside the root cortex. Examples,
Heterodera spp. and *Meloidogyne* spp.
b)
Semiendoparasitic nematodes
The anterior part of nematode, head and neck being perm
anently fixed in the
cortex and the posterior part extends free into the soil. Examples,
Rotylenchulus
reniformis
and
Tylenchulus semipenetrans
.
c) Ectoparasitic nematodes
These nematodes live freely in the soil and move closely or on the root surface, feed intermittently on the epidermis and root hair near the root tip. They are two types,
1) Migratory ectoparasites
These nematodes spend their entire life cycle free in the soil, feeding externally on the host plants, deposit eggs in soil. When the 145 roots are disturbed they detach themselves. Examples,
Criconemoides spp., *Paratylenchus* spp. and *s* spp., etc.

2) ***Sedentory ectoparasites***
In this type of parasitism the attachment of nematode to the root system is permanent but for this, it is similar to the previous one. Examples, *Hemicycliophora arenaria* and *Trichodorus* spp., etc.
1)
Rootknot Nematode,
***Meloidogyne* spp.**
Systematic Position
Order
-
Tylenchida
Sub order
-
Tylenchina
Super family
-
Tylenchoidea
Family
-
Heteroderidae
Sub family
-
Meloidogyninae
Genus
-
Meloidogyne
Species
-
i) *incognita*
ii) *javanica*
iii) *arenaria*
iv) *hapla*

Parasitism & Habitat:

i) Females and III & IV stage of larvae are Sedentory endoparasites.
ii) Males and II stage larvae are migratory.
Morphological characters:
i) Body
- Elongate larvae and male Typically saccate, spheroid with a distinct neck in females.

ii) Stylet
- In males, Strong with rounded knob & In females, more slender than males.
iii) Oesophagous
– With large median bulb followed by short isthumus.

iv) Excretory pore
- Often seen with part of excretory tube in the area between posterior part of stylet knobs and opposite to median bulb.
v) Vulvas & anus
- Females typically opposite to neck and surrounded by a pattern of fine lines like human fingerprint.(Perennial pattern)

Spicule
-Very near the terminus of males Bursa is absent.
- Yellowing of leaves
- Stunted growth
- Reduced vigor 146

- Reduced size & number of fruits
- Gall formation
- **Multinucleate cell**
– Giant cell (Nurse cell)
- **Hypertrophy** -Enlargement of cell
-
Hyperplasia -Multiplication of cell

Control:
- Two to three deep Ploughing
- Rotation with cereal crops
- Apply carbofuron (Furdan 3G) @ 7 g/m
Resistant varieties of Tomato

Symptoms Caused by Nematodes
Most of the plant parasitic nematodes affect the root portion of plants
except
Anguina spp,
Aphelenchus spp,
Aphelenchoides spp,
Ditylenchus spp,
Rhadinaphelenchus cocophilus
and *Bursaphelenchus xylophilus.*
Nematode suck the sap of the plants with the help of stylet and causes leaf discolouration, stunted growth, reduced leaf size and fruits and lesions on roots, galls, reduced root system and finally wilting.
Symptoms of nematode disease can be classified as
A)
Symptoms produced by above ground feeder nematodes
B)
Symptoms produced by below ground feeder nematodes
A)
Symptoms produced by above ground feeder nematodes
i) Dead or devitalized buds
–
Nematode infection kills growing buds *e.g. Aphelenchoides fragariae* on strawberry.
ii) Crinkled stems and foliage
- *e.g.* Wheat gall nematode, *Anguina tritici* Ulfa disease of rice, *Ditylenchus angustus.*
iii) Seed galls
– *e.g.* Wheat gall nematode, *Anguina tritici* larva enter into the flower primordium and develops in to a gall.
iv) Necrosis & discolouration
– Red ring disease of coconut, *Rhadinaphelenchus cocophilus*
. Due to the infestation, red coloured circular area appear in the trunk of infested palm.
v) Leaf lesions
- Symptom on broad
-leafed foliage plants. *e.g.* Chrysanthemum foliar nematode, *Aphelenchoides ritzemabosi*
vi) Twisting of leaves and stem: *e.g.* In onion basal leaves become twisted when infested with *Ditylenchus dipsaci.*
vii) Leaf discolouration:
The leaf tip become white in rice due to rice white tip nematode *Aphelenchoides besseyi.*

B) Symptoms produced by below ground feeder nematodes
The nematode infest and feed on root portion and exhibit symptoms on below ground plant part as well as on the above ground plants parts and they are classified as
I) Above ground symptoms
II) Below ground symptoms

I) Above ground symptoms:

i. Stunting:
Reduced plant growth and the plant cannot able to withstand in adverse conditions. Patches of stunted plants appears in the field.
e.g. Heterodera avenae
– Molya disease in wheat & barley.
Globodera
rostochiensis
– Golden nematode in potato 154
ii. Discolouration of foliage: Also due to nutritional deficiency
e.g.
Root lesion nematode, *Pratylenchus coffeae* White tip nematode,
Aphelenchoids besseyi Citrus nematode, *Tylenchulus semipenetrans*

iii. Wilting: *e.g.* Root-knot nematodes, *Meloidogyne spp*

iv. Decline and die back: eg. In banana decline and die back are caused by *Radopholus similis.*

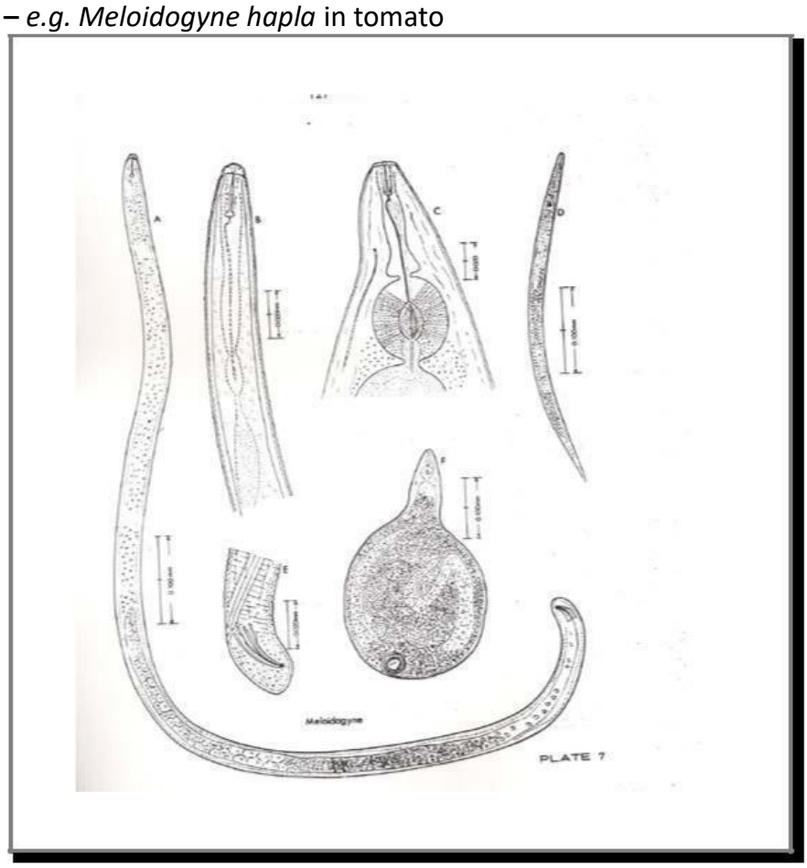
II) Below ground symptoms:
i. Root galling:
e.g. ***Meloidogyne spp***
.
–
Characteristic galls on host roots
Nacobbus spp
–
Larger galls on beet & tomato
Ditylenchus radicicola
–
Small galls on cereals.
Hemicycliophora arenaria
–
Galling on lemon roots
Xiphinema diversicaudatum
-
Galling on roses
ii) Reduced root system:
Due to nematode feeding the root tip growth is arrested and the root produced branches. This may be of various kinds such as coarse root, stubby root and curly root.
a) Stubby roots
– Stubby branches or rootlets arranged in cluster eg. *Trichodorus christiei* on corn

b) Coarse root
– Lateral roots stopped growth with no branches *e.g. Belonolaimus longicaudatus* on corn.

c) Curly root
–
The nematode retard the elongation of roots and cause curling of roots known as
'Fish hook' symptom. Eg. Injury caused by *Xiphinema* spp.

iii)
Root lesions
– Necrotic lesions *e.g. Pratylenchus spp* (soybean), *Radopholus similis* (citrus & banana), *Helicotylenchus multicinctus* (banana)
iv) Rotting
– Nematode + Micro
-organisms.
e.g. Ditylenchus destructor
– potato rot.
v) Excessive root branching

– *e.g. Meloidogyne hapla* in tomato



3.2 Symptoms caused by nematodes

- Nematode infections of plants result in the appearance of symptoms on roots as well as on the aboveground parts of plants.
- Root symptoms may appear as root lesions, root knots or root galls.
- Excessive root branching, injured root tips, and, when nematode infections are accompanied by plant pathogenic or saprophytic bacteria and fungi, as root rots
- Some species of nematodes attacks upper part of the plant and cause twisting or distortion of leaves and stems, and abnormal development of the floral parts.
- Certain nematodes attack cereals or grasses and form galls full of nematodes in place of seed.

3.1 How nematodes affect plants

Most of the damage by nematodes seem to be caused by a secretion of saliva injected into the plants while the nematodes are feeding.

Some nematode species are rapid feeders. They puncture a cell wall, inject saliva into the cell, withdraw part of the cell contents, and move on within a few seconds.

Others feed much more slowly and may remain at the same puncture for several hours or days. These, as well as the females of species that become established in or on roots permanently, inject saliva intermittently as long as they are feeding.

The feeding process causes the affected plant cells to react, resulting in dead or devitalized root tips and buds, lesion formation and tissue breakdown, swellings and galls of various kinds, and crinkled and distorted stems and foliage.

Other results of infection causes abnormal cell enlargement **(hypertrophy),** by suppression of cell divisions, or by stimulation of cell division proceeding in a controlled manner and resulting in the formation of galls or of large numbers of lateral roots at or near the points of infection.

Plant diseases caused by nematodes are complex. Root-feeding species of nematodes often decrease the ability of plants to take up water and nutrients from soil and thus cause symptoms of water and nutrient deficiencies in the aboveground parts of plants.

It is also important to note that some relationships between nematodes and other pathogens like fungi and bacteria do exists and enhance chances of causing disease.

3.2 Control of nematodes

- Control is usually attempted through cultural practices, such as use of clean planting stock, crop rotation, fallow, and cover crops; through biological control
- by means of physical agents, such as tillage, heat, including solarization, and flooding;
- control with chemicals, such as various types of fumigants
- however, best control approach is a combination of several methods (Integrated Pest Management- IPM) for effective control of nematode diseases of plants.

LECTURE 7.

1.0 METHODS OF DETECTION OF PLANT DISEASES

Detection and identification of diseases in crops could be realized via both **direct** and **indirect methods**.

Direct detection of diseases includes molecular and serological methods that could be used to analyze large numbers of samples that needs to be analyzed. In these methods, the disease causing pathogens such as bacteria, fungi and viruses are directly detected to provide accurate identification of the disease/pathogen.

On the other hand, **indirect methods** identify the plant diseases through various parameters such as morphological change, temperature change, transpiration rate change and volatile organic compounds released by infected plants.

1.1 Direct Detection Methods

Some of the direct detection methods are as follows:-

i. Polymerase Chain Reactions (PCR)

Polymerase chain reaction (PCR) is a common molecular biology technique that enables researchers to make multiple copies of a specific region of DNA. PCR is efficient, rapid and can amplify DNA or RNA sequences from various sources. Once the DNA has been sufficiently amplified, the resulting product can be sequenced, analyzed by gel electrophoresis, or cloned.

PCR was initially used for highly specific detection of diseases caused by bacteria and viruses but now it has been widely used for the detection of plant pathogens as well.

Some advantages of PRC are that, it's a common technology, portable, easy to operate but has some limitations like; its effectiveness is subjected to DNA extraction, inhibitors, polymerase activity, and concentration of PCR buffer.

ii. Fluorescence In-Situ Hybridization (FISH)

This is another type of molecular detection technique which is applied for bacterial detection in combination with microscopy and hybridization of DNA probes and target gene from plant samples.

Due to the presence of pathogen-specific ribosomal RNA (rRNA) sequences in plants, recognizing this specific information by FISH can help detect the pathogen infections in plants. In addition to bacterial pathogens, FISH could also be used to detect fungi and viruses and other endosymbiotic bacteria that infect the plant. One of its advantges is that it has very high sensitivity.

iii. Enzyme-Linked Immunosorbent Assay (ELISA)

The enzyme-linked immunosorbent assay (ELISA) is another molecular method for identification of diseases based on **antibodies** and **color change in the assay**. In this method, the target epitopes (antigens) from the viruses, bacteria and fungi are made to specifically bind with antibodies conjugated to an enzyme.

The detection can be visualized based on color changes resulting from the interaction between the substrate and the immobilized enzyme.

Some **advantages** of this method include low cost, visual color change can be used for detection while the greatest disadvantage is that it has low sensitivity for bacteria.

1.2 Indirect detection methods

These are methods based on the features being expressed by the plants such as leaf size, using optical sensors to detect biotic and abiotic stress in plants etc.

7.2 REMOTE SENSING

Remote sensing is estimating an object/phenomenon without being in physical contact with it. Remote sensing is a science/art that permits us to obtain information about an object/a phenomenon through analysis of data obtained through sensory devices without being in physical contact with that object.

Objectives of remote sensing in plant Pathology

1. Assessment of disease over a vast area
2. To know the relationship of diseases and environment
3. To know the origin and development of epidemics
4. Quantitative assessment of the disease

Remote sensing techniques of importance to Plant Pathology

1. Aerial photography and 2. Satellite remote sensing

1. Aerial photography: Aerial photography can detect objects on land over a larger area.**Colwell** (1956) first used remote sensing technique for monitoring stem rust of wheat. He showed that panchromatic colour and especially infrared aerial photography could be used to detect rusts and viral diseases of small grains and certain diseases of citrus. Later, infrared photography was used in England for late blight of potato.

The key to distinguish diseased and healthy parts of a crop is to use appropriate film or filter combinations. The main film types used are panchromatic, infrared, normal colour and colour infrared. The **infrared** films are preferred because of their superior sensitivity to visible light and to near infrared wavelengths of radiation (700-900 mμ). The colour infrared or Ektachrome Aero Infrared (Camouflage Detection Film) is superior as it can show the difference between diseased and healthy patches of plants in colour. Thehealthy foliage is highly reflective to the infrared wavelengths and appears red on this film whereas blighted or diseased foliage has low infrared reflectance and does notappear red in the photograph.

2. Satellite Imaging

Weather satellites

Often cyclones create heavy clouds with rains and an anti-cyclone creates a cloudless sky. All these can be effectively monitored by weather satellites. Sequential pictures show the movement of these systems before they arrive in an area. Therefore by monitoring epidemic favouring systems using a satellite, the disease occurrence on the field can be monitored. Ex: The spread and deposition of stem rust pathogen of wheat is influenced by definite synoptic weather conditions called **Indian stem rust rules**.

Earth resources technology satellites (LANDSAT, 1972, USA)

LANDSAT covers the entire globe every 18 days scanning the same area at a fixed time. The scanned data is compared for any major differences happened within 18 days.Nagarajan utilized LANDSAT infrared spectral bands 6 (0.7-0.8μm) and 7 (0.8-1.1μm)to differentiate healthy wheat crop of India and severe yellow rust affected crop of Pakistan.

Examples: Coconut root rot and wilt, black stem rust of wheat, citrus canker

Advantages of Remote sensing

1. Reveals pattern of disease incidence, intensity and development over large area
2. Data generated by remote sensing is amenable to multidisciplinary approach
3. Gives synoptic view of large areas
4. Data generated is on a permanent scale and is unbiased
5. Data acquisition is fast compared to traditional methods and data analyzed is effectively utilized
6. Satellite data (ERTS) obtains information of an area periodically so that the information can be updated.
7. It frequently poses questions for ground investigators which cannot be generated byground parties.

Lecture 8

INTEGRATED PEST DISEASE MANAGEMENT (IPDM)

1.0 INTRODUCTION

IPDM involve management systems which utilize compatible combinations of all the available techniques to keep the pathogen population below the economic threshold level (ETL) which would not result in economically unacceptable damage to the crop. IPDM is based on five principles of plant disease management and integrates multidisciplinary approaches for the management of plant diseases.

Main components of IPDM:
1. Cultural practices
2. Regulatory measures (quarantine)
3. Chemical methods
4. Biological methods
5. Physical methods
6. Genetic engineering

Main strategies of IPDM:
1. Need based application of pesticides
2. Encouragement and enhancement of biocontrol agents
3. Use of resistant or tolerant cultivars of plants
4. Modification of cultural practices
5. Use of any other strategies that interrupts host-pathogen interactions

Advantages of IPDM:
1. Avoids chemical pollution of soil, water, air and food products
2. Avoids development of resistance in the plant pathogens against fungicides
3. It is an eco-friendly strategy for management of plant diseases
4. It is an economically feasible approach
5. It is a multipronged strategy for efficient management of plant diseases

Therefore, IPDM utilizes all suitable strategies in a compatible manner to reduce andmaintain pathogen populations at levels below those causing economic losses.

Rice diseases and IPDM:
Fungal diseases
1. Blast: Foliar disease and the pathogen survives on collateral hosts
2. Brown spot of rice – Seed borne and a foliar disease
3. Sheath rot, sheath blight, foot rot and stem rot – Soil borne diseases
4. False smut – seed borne disease

Bacterial diseases: Bacterial leaf blight and bacterial leaf streak – Seed borne and survives on collateral hosts and weeds

Viral or Phytoplasmal diseases – Rice tungro virus, Rice yellow dwarf – Survives onweeds and dissemination is by insect vectors

IPDM strategy in rice:
1. Selection of healthy seed
2. Selection of resistant cultivars
3. Removal and destruction of collateral hosts
4. Balanced fertilization
5. Rouging of diseased plants
6. Seed treatment with carbendazim or tricyclazole at 2g/Kg seed
7. Need based foliar application of carbendazim@0.1% or Tricyclazole@0.06% for themanagement of blast.

8. Need based foliar application of validamycin for the management of sheath blight andsheath rot.
9. Soil application of carbofuran granules or foliar spray of any systemic fungicide isfollowed to manage insect vectors, thereby decreasing the spread of viral diseases.

2.0 INTEGRATED PEST MANAGEMENT (IPM)

1.0 INTRODUCTION

What is IPM?

Integrated Pest Management (IPM) is an effective and environmentally sensitive approach to pest management that relies on a combination of common-sense practices. IPM programs use current, comprehensive information on the life cycles of pests and their interaction with the environment. This information, in combination with available pest control methods, is used to manage pest damage by the most economical means, and with the least possible hazard to people, property, and the environment.

The IPM approach can be applied to both agricultural and non-agricultural settings. IPM takes advantage of all appropriate pest management options including, but not limited to, the good use of pesticides. In contrast, organic food production applies many of the same concepts as IPM but limits the use of pesticides to those that are produced from natural sources, as opposed to synthetic chemicals.

How do IPM programs work?

IPM is not a single pest control method but, rather, a series of pest management evaluations, decisions and controls. In practicing IPM, growers who are aware of the potential for pest infestation follow a four-tiered approach. **The four steps include:**

i. Set Action Thresholds

Before taking any pest control action, IPM first sets an action threshold, a point at which pest populations or environmental conditions indicate that pest control action must be taken. Sighting a single pest does not always mean control is needed. The level at which pests will become an **economic threat** is critical to guide future pest control decisions.

ii. Monitor and Identify Pests

Not all insects, weeds, and other living organisms require control. Many organisms are innocuous, and some are even beneficial. IPM programs work to monitor for pests and identify them accurately, so that appropriate control decisions can be made in conjunction with action thresholds. This monitoring and identification removes the possibility that pesticides will be used when they are not really needed or that the wrong kind of pesticide will be used.

iii. Prevention

As a first line of pest control, IPM programs work to manage the crop, lawn, or indoor space to prevent pests from becoming a threat. In an agricultural crop, this may mean using cultural methods, such as rotating between different crops, selecting pest-resistant varieties, and planting pest-free rootstock. These control methods can be very effective and cost-efficient and present little to no risk to people or the environment.

iv. Control

Once monitoring, identification, and action thresholds indicate that pest control is required, and preventive methods are no longer effective or available, IPM programs then evaluate the proper control method both for effectiveness and risk. Effective, less risky pest controls are chosen first,

including highly targeted chemicals, such as pheromones to disrupt pest mating, or mechanical control, such as trapping or weeding. If further monitoring, identifications and action thresholds indicate that less risky controls are not working, then additional pest control methods would be employed, such as targeted spraying of pesticides. Broadcast spraying of non-specific pesticides is a last resort

IPM takes advantage of all appropriate pest management options including, but not limited to, the use of pesticides. In contrast, organic food production applies many of the same concepts as IPM but limits the use of pesticides to those that are produced from natural sources, as opposed to synthetic chemicals.

Before taking any pest control action, IPM first sets an action threshold, a point at which pest populations or environmental conditions indicate that pest control action must be taken. Sighting a single pest does not always mean control is needed. The level at which pests will either become an economic threat is critical to guide future pest control decisions.

LECTURE 9

1.0 PLANT TISSUE CULTURE: *In vitro* culture of plant cells, tissues as well as organs. **Totipotency** is the ability of a plant cell to perform all the functions of development which are characteristic of zygote, i.e., its ability to develop in to a complete plant.

IMPORTANT TISSUE CULTURE TECHNIQUES OF IMPORTANCE TO PLANT PATHOLOGY:

1. Meristem tip culture
2. Protoplast culture

A. Production of virus free plants through plant tissue culture:

Meristem tip culture: Cultivation of axillary or apical meristems, particularly of shoot apical meristem, is known as meristem culture.

1. Explant: the explant must consist of the meristematic dome of cells together with atleast one leaf primordial. Meristem tips varying in size from 0.1 to 2.0 mm in diameter (usually 0.3-1.5 mm) can be used for meristem tip culture. The infected parent plant or organ of the plant from which explant is excised is generally subjected to thermotherapy in a temperature controlled cabinet at 30^0C to 40^0C for six to twelve weeks to inactivate the virus.

2. Culture initiation on suitable medium: In general suitable medium is used for culture using this method . But for some species, a much lower salt concentration may be adequate or even necessary since the high salt concentration of some other medium may be deleterious or even toxic. Culture initiation consists of surface sterilization of explants and establishing them *in vitro* on culture medium. Culture initiation often involves anti-metabolite chemicals such as ribavirin (virazole) in the tissue culture medium.

3. Shoot multiplication: After 2-3 weeks, the cultures are transferred to a shoot multiplication medium designed to promote axillary branching. This medium generally contains cytokinins, either alone or in combination with an auxin. Higher concentration of cytokinins induces adventitious buds. During culture initiation and shoot multiplication phases, the cultures are generally kept at 25^0C.

4. Rooting of shoots: In general, the rooting medium has low salt (1/2 or even ¼ salts of MS medium) and reduced sugar levels. But in most species, 0.1-1 mg/l Naphthalene Acetic Acid (NAA) or Indole-3-Butyric acid (IBA) is required for rooting. Rooting takes about 10-15 days depending on species.

5. Transfer of plantlets to soil: Rooted shoots are removed from the medium, agar sticking to roots is washed with tap water, and they are transplanted into plastic cups containing a suitable potting mix. Plants are kept in high (>90%) humidity and initially low light intensities. The humidity is generally decreased to the ambient level after about 7-15 days, and the light intensity is increased. The plants are finally exposed to greenhouse conditions (**hardening**).

6. Indexing, clone selection and stock maintenance: Virus indexing is done several

times during first year and the virus free plantlet is used as a nuclear stock material for commercial multiplication. Virus indexing is generally made by Enzyme Linked Immuno-Sorbent Assay (ELISA) or Immuno Sorbent Electron Microscopy (ISEM).

B. Protoplast culture: Fungal protoplasts are important tools in physiological and genetic research. Interspecific, intraspecific and intrageneric hybridization could be done by this technique for strain improvement of biocontrol agents to enhance the biocontrol potential for the management of pathogenic fungi. Isolation and self-fusion of protoplasts were achieved in *Trichoderma harzianum* and *T. viride*.

Steps in protoplast fusion:

1. Isolation of protoplasts is achieved by treating cells with a suitable mixture of cell wall degrading enzymes.
2. The pH of enzyme solution is adjusted between 4.7 and 6.0 and temperature is kept around 25-30^{0}C. The osmotic concentration of enzyme mixture and of subsequent media is elevated to stabilize the protoplasts and to prevent them from bursting. Usually, 50-100 m mol/l $CaCl_2$ is added to the osmoticum as it improves plasma membrane stability.
3. The protoplasts of different strains are treated with 28-50% **Poly Ethylene Glycol** (**fusogen**) for 15-30 min followed by gradual washing of the protoplasts to remove PEG. The washing medium may be alkaline and contain high calcium ion concentration (50 m mol/l). Protoplast fusion occurs during washing step.
4. Selection of hybrid cells and culturing on suitable medium.

Gene cloning/ Recombinant DNA technology / Genetic engineering:

Integration of specific fragment of foreign DNA into a cell through a suitable vector in such a way that the inserted DNA replicate independently and transferred to progenies as a result of cell division.

Recombinant DNA molecule is a vector into which the desired DNA fragment has been inserted to enable its cloning in an appropriate host. Recombinant DNA molecule is produced by joining together two or more DNA segments usually originated from different organisms.

Steps in gene cloning:

1. Identification and isolation of the desired gene or DNA fragment to be cloned (Restriction digestion and electrophoresis)
2. Insertion of the isolated gene in a suitable vector (ligation)
3. Introduction of this vector into a suitable organism or cell called host (transformation)
4. Selection of transformed host cells (selectable markers)
5. Multiplication / integration followed by expression of the introduced gene in the host

Enzymes involved: Restriction endonucleases, DNA ligases, DNA polymerases, RNA polymerases and reverse transcriptases.

Vectors used in gene cloning: A vector is a DNA molecule that has the ability to replicate in an appropriate host cell, and into which the DNA fragment to tbe cloned (called DNA insert) is integrated for cloning. Ex: Tumor inducing (Ti) plasmid of *Agrobacterium tumefaciens*, pBR322, Bacteriophages, cosmid vectors (derived from phage λ).

Principles and methods of plant disease management

Plant Disease Management

The word 'control' is a complete term where permanent 'control' of a disease is rarely achieved whereas, 'management' of a disease is a continuous process and is more practical in influencing adverse affect caused by a disease. Disease management requires a detail understanding of all aspects of crop production, economics, environmental, cultural, genetics and epidemiological information upon which the management decisions are made.

A, **Principles of plant disease management:**

There is six basic concept or principles or objectives lying under plant disease management.

1. **Avoidance of the pathogen**

: Occurrence of a disease can be avoided by planting/sowing a crop at times when, or in areas where, inoculum remain ineffective/inactive due to environmental conditions, or is rare or absent.

2. **Exclusion of the pathogen**

: This can be achieved by preventing the inoculum from entering or establishing in a field or area when it does not exist. Legislative measures like quarantine regulations are needed to be strictly applied to prevent spread of a disease.

3. **Eradication of the pathogen**

: It includes reducing, inactivating, eliminating or destroying inoculum at the source, either form a region or from an individual plant (rouging) in which it is already established.

4. **Protection of the host**

: Host plants can be protected by creating a toxin barrier on the host surface by the application of chemicals.

5. **Disease resistance**

: Preventing infection or reducing the effect of infection of the pathogen through the use of resistance host which is developed by genetic manipulation or by chemotherapy.

6. **Therapy**

: Reducing severity of a disease in an infected individual.

The first five principles are prophylactic (preventive) procedure and the last one is curative.

B. Methods of plant disease management

1**Avoidance of the pathogen:**

i. Choice of geographical area

ii. Selection of a field

iii. Adjustment of time of sowing

iv. Use of disease escaping varieties

v. Use of pathogenfree seed and planting material

vi. Modification of cultural practice